

Big Data Analysis of Online Retail Transactions Using Apache Spark

**IT 462 – Big Data Systems
Group 5**

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Problem and Motivation

Context



Retail businesses collect large transactional data but often lack clear insight into customer behavior, loyalty, and inactivity.

Problem



Identify meaningful customer segments and potential churn-risk customers using RFM features from unlabeled retail transaction data.

Target User



Marketing teams, business managers, and customer retention analysts.

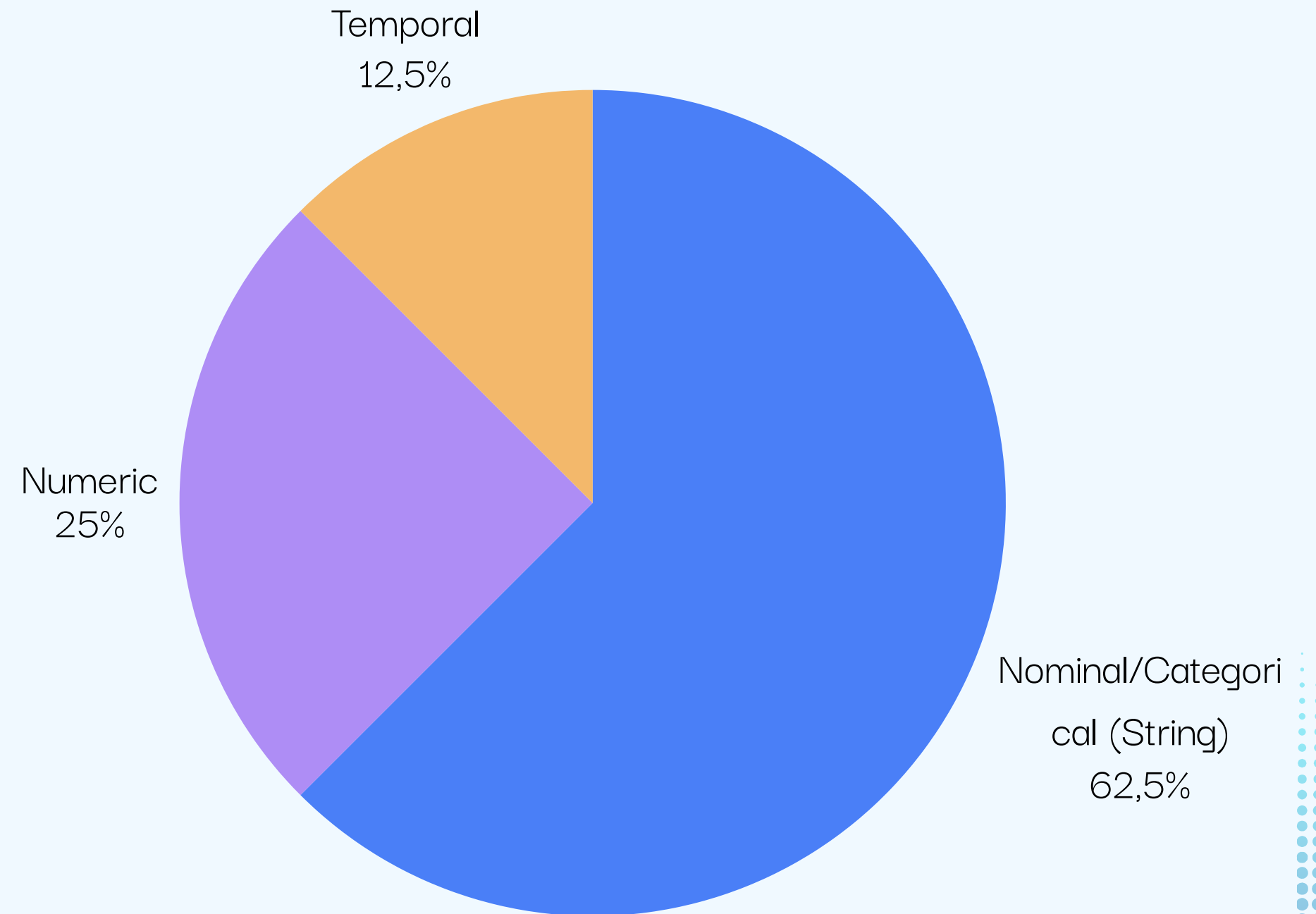
Why it matters



The results help businesses detect valuable and at-risk customers, compare purchasing patterns, and design targeted retention strategies.

Dataset Overview

- Online Retail II, from Kaggle
- 1,067,371 rows and 8 columns.
- E-commerce and Online Retail Analytics



Data Quality & Preprocessing

Main Data Quality Issues

- Missing values were found mainly in CustomerID and Description.
- Invalid transaction records were detected, such as Quantity $\leq$ 0 and UnitPrice $\leq$ 0.
- Duplicate records and outliers in Quantity .

Key Preprocessing Steps

- Removing rows without CustomerID and replacing missing Description with “Unknown.”
- Removed invalid and duplicate transactions to improve data quality.
- Selected relevant features and transformed the data into customer-level RFM features for analysis.

Issue	Before	After
Missing CustomerID rows	243,007	0
Missing Description	4382	Replace with “Unknown”
Invalid Quantity/Price	18,815	0
Duplicate records	26,124	0
Outliers	51,119	0
Final dataset size	1,067,371	728,306

RDD Exploration: Churn Candidate Identification

Operations:

map → Extract customer RFM features from each row

filter → Identify customers with high churn risk

take → Retrieve the top inactive customers

Operations:

```
val customerFeatures = rows.map(row =>
  (row(0), row(1), row(3).toInt, row(4).toInt, row(5).toDouble)
)

val churnRDD = customerFeatures.filter {
  case (_, _, recency, frequency, _) =>
    recency > 90 && frequency < 5
}
```

Sample Result:

CustomerID	RecencyDays	Frequency
13526	738	2
17056	738	1
12636	738	1

RDD Exploration: Country-Level Aggregation

Operations:

map → Convert customer records into country-based pairs

reduceByKey → Aggregate churn metrics per country

mapValues → Compute average spending and churn ratio

Operations:

```
val countryRDD = customerRDD.map(x =>
  (x._1, (1, x._3, x._4)))

val result = countryRDD.reduceByKey((a,b) =>
  (a._1+b._1, a._2+b._2, a._3+b._3))
```

Sample Result:

Country	High-Risk	Churn Ratio
Australia	5	333
Brazil	1	500
Canada	2	500

SQL Analytics Highlights

Important Queries



- **Country Value Profile**

Identified countries with the highest customer spending and frequency.

- **High-Value Inactive Customers**

Detected customers with high spending but long inactivity periods.

- **Churn Percentage by Country**

Measured churn-risk rates across different countries.

Example Result

High-Value Inactive Customers

CustomerID	Country	RecencyDays	Frequency	Monetary	risk_segment
13284	United Kingdom	731	1	966.66	Critical Risk
13768	United Kingdom	726	2	875.97	Critical Risk
14061	United Kingdom	724	2	553.43	Critical Risk

Brief Interpretation

SQL analysis revealed clear behavioral differences between active and inactive customers. These findings supported the use of RFM features in clustering to identify loyal customers and churn-risk segments.



ML Approach

Problem Type



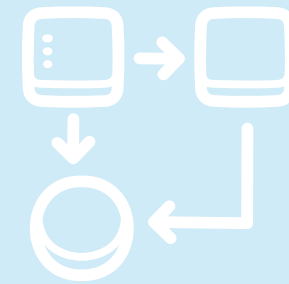
Unsupervised learning
(Clustering)

Features Used



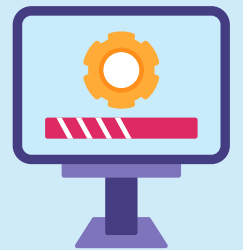
- RecencyDays
- Frequency
- Monetary (RFM Model)

Models Used



- **K-Means**
simple and efficient
- **Gaussian Mixture Model(GMM)**
probabilistic clustering

Training Setup



- Train/Test Split: **70% / 30%**
- Random Seed: **42**
- Evaluation Metric: **Silhouette Score**

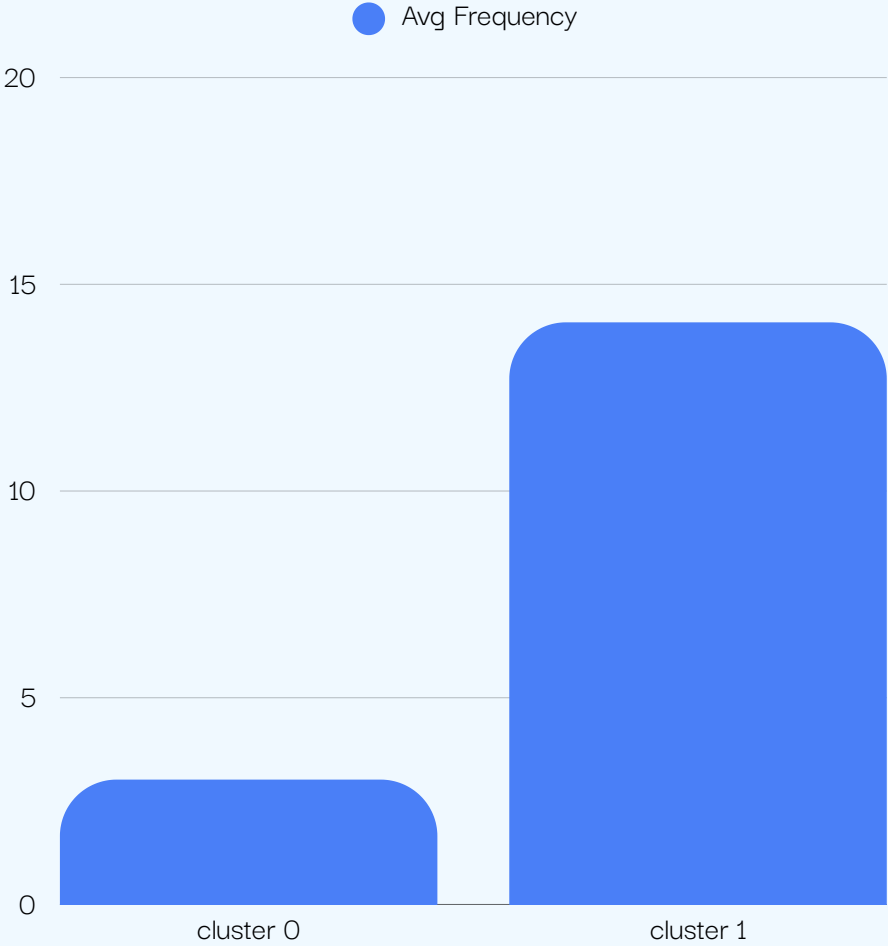
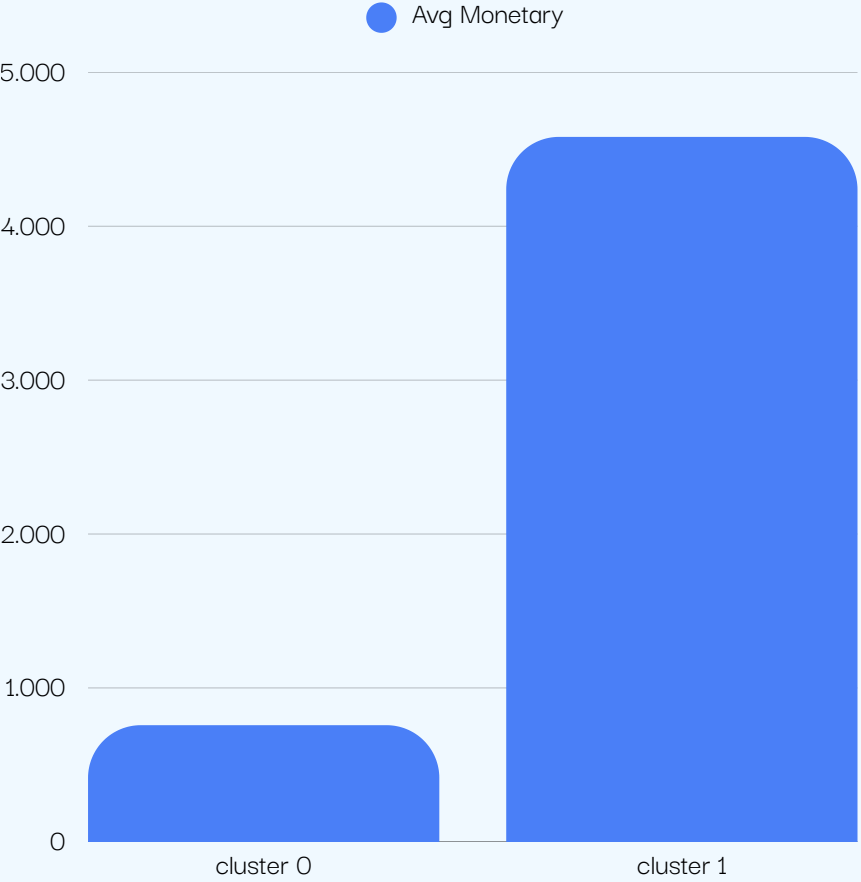
Results & Model Performance

Results

Model	Best K	Silhouette Score
K-Means (Baseline)	2	0,7232
GMM	2	0,6419

Results & Model Performance

Average RFM Values by Cluster



Interpretation & Insights

- K-Means achieved the best clustering performance, with a silhouette score of 0.7232, compared to 0.6419 for GMM.
- The model separated customers into two main groups:
 - Loyal high-value customers: low recency, high frequency, and high monetary value.
 - Inactive low-value customers: high recency, low frequency, and lower spending.
- RecencyDays and Frequency were the most important behavioral indicators for distinguishing active customers from churn-risk customers.
- The results can help businesses target inactive customers with retention campaigns and prioritize loyal high-value customers with personalized offers.

Limitations & Future Work

Limitations:



The analysis depends on RFM features



historical from 2009–2011, so it may not fully reflect current behavior.



Some countries have a very small number of customers

Future Work:

- Add more data sources, such as customer demographics, product categories,.
- Test more clustering models, such as DBSCAN and hierarchical clustering.
- Improve the model through deeper feature engineering, hyperparameter tuning.
- Build an interactive dashboard to monitor customer segments, churn-risk customers.

THANK YOU

Any Question?

